HEMALATHA YALAMANCHI

AI/ML Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience designing and deploying production-grade AI systems for intelligent document processing, automation, and data-driven workflows. Expertise spans OCR, NLP, Large Language Models (LLMs), Vision Language Models (VLMs), and cloud-native architectures across Azure and GCP. Proven track record building secure, scalable pipelines that integrate AI services, REST APIs, and modern data platforms — driving measurable efficiency gains and cost savings. Strong foundation in MLOps, CI/CD automation, and cross-functional collaboration with engineering, compliance, and operations teams.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash, Java
AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, XGBoost, Hugging Face Transformers
NLP & Document AI: spaCy, NLTK, Tesseract OCR, Vision Language Models (VLMs), Named Entity Recognition, Tokenization
LLMs & GenAI: OpenAI API, Prompt Engineering, RAG, Embeddings, LLM-based Classification & Parsing
Cloud Platforms: Azure (Key Vault, App Services, Cognitive Services, Managed Identities, DevOps), GCP (Cloud Storage), Render Cloud
Databases: PostgreSQL, MongoDB, MySQL, Snowflake
Backend & APIs: REST APIs, Microservices, API Security, UUID-based Tracking
DevOps & MLOps: CI/CD Pipelines, Docker, Git/GitHub, Linux (RHEL 8/9), Monitoring & Alerting
Visualization & BI: Power BI, Tableau
Methodologies: Agile/Scrum, Test Automation, Unit Testing, Compliance & Documentation

PROFESSIONAL EXPERIENCE

AI/ML Engineer — Tax Relief Advocates, Irvine, CA *Oct 2025 – Present*

- Designed and deployed end-to-end AI automation workflows for unstructured document processing, eliminating manual handling and significantly improving operational throughput.
- Automated document processing pipelines previously performed manually, contributing to an estimated **$7M+ in projected annual cost savings** through efficiency gains and reduced labor dependency.
- Built OCR + NLP + **Vision Language Model (VLM)** pipelines to extract structured fields from scanned documents, achieving **>95% extraction accuracy** across diverse document layouts.
- Developed **LLM-powered document classification and parsing systems**, reducing manual review effort by **40%+** while maintaining production-grade reliability.
- Engineered cloud-to-database migration pipelines that converted unstructured files from Google Drive into normalized PostgreSQL and MongoDB schemas for downstream analytics and automation.
- Designed and implemented secure REST APIs for document ingestion, processing, and downstream service integration with proper authentication and rate-limiting.
- Built enterprise-grade security architecture using **Azure Key Vault** for secrets management, **Managed Identities** for service-to-service authentication, and IP-restricted access controls for sensitive systems.
- Implemented **UUID-based tracking system** for end-to-end document traceability across distributed pipelines, enabling reliable auditing and debugging.
- Deployed scalable microservices on **Render Cloud and Azure App Services**, utilizing Google Cloud Storage (GCS) for object storage and retrieval at scale.
- Designed and maintained PostgreSQL and MongoDB databases for structured and semi-structured data, optimizing schema design, indexing, and query performance.
- Implemented case-ID matching and entity resolution logic to handle edge cases, improving record-linking accuracy and reducing data duplication.
- Built proactive monitoring, retry mechanisms, and alerting systems, ensuring high pipeline uptime and rapid incident response.
- Collaborated with compliance, operations, and engineering teams to validate model outputs and ensure regulatory and audit-readiness alignment.

Data Scientist — Srivina Tech (Remote)*Jun 2025 – Oct 2025*

- Designed and deployed OCR + NLP pipelines using Python and TensorFlow to extract structured data from scanned PDFs, improving extraction accuracy by **30%**.
- Developed AI workflows integrating Azure AI Cognitive Services and OpenAI APIs for text classification, document parsing, and knowledge extraction.
- Configured and deployed ML workloads on RedHat Enterprise Linux (RHEL 9), optimizing resource utilization for production workloads.
- Built and validated automated CI/CD pipelines with Azure DevOps for AI models, reducing deployment cycle time by **25%**.
- Partnered with subject-matter experts to validate model outputs and align with compliance standards, achieving >95% precision in extracted fields.

Data Scientist — REWBIX Inc (Remote)*Jun 2024 – Jun 2025*

- Implemented document parsing ML models in Python using scikit-learn and TensorFlow for structured extraction from financial and business reports.
- Enhanced parsing pipelines with NLP techniques (tokenization, entity recognition, regex-based extraction), persisting structured outputs to MySQL databases.
- Automated testing and validation frameworks to ensure compliance-ready AI workflows with **98% test coverage**.
- Developed Tableau dashboards and automated KPI reporting from parsed datasets, improving stakeholder decision-making speed by **18%**.
- Collaborated in Agile teams using JIRA, delivering AI-driven solutions with 98% on-time release.

Teaching Assistant, Predictive Modeling — Saint Louis University, MO*Aug 2024 – Dec 2024*

- Mentored graduate students in Python NLP, predictive modeling, and ML workflows using scikit-learn and Jupyter.
- Supported projects involving structured text parsing and classification, guiding students on tokenization and data-cleaning methods.
- Automated grading via Linux-based Python scripts, cutting turnaround time by 25%.

KEY PROJECTS

Intelligent Document Parser — Capstone Project, SLU*2024*

- Built an OCR + NLP pipeline using Tesseract OCR and spaCy to parse structured fields from standardized forms, achieving **92% field-extraction accuracy**.
- Stored results in a normalized MySQL schema and integrated outputs into a downstream automation pipeline as a proof-of-concept for end-to-end document workflows.

Network Latency Predictor — Personal Project*Jan 2025 – Mar 2025*

- Designed an ML model in scikit-learn to predict network latency with **90% accuracy**, supporting proactive performance optimization.
- Automated KPI reporting pipeline and deployed the model on Linux with Power BI dashboard integration.

EDUCATION

Master of Science, Data Science — Saint Louis University, MO*Dec 2024*

Relevant Coursework: Machine Learning, Predictive Modeling, Database Systems, Data Mining, Natural Language Processing, Linux System Administration

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — 2024
- **TensorFlow Developer Certificate** — DataCamp, 2025
- **Data Science with Python & R** — DataCamp, 2024
- **Project Management & Digital Strategy** — SLU Excelerate, 2023